

Muhammad Raka Aiko Pramudityo

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SUMMARY

Undergraduate Informatics student at Universitas Teknologi Digital Indonesia with an interest in Full Stack Development, spanning user interface development (frontend) through to backend systems and databases. Experienced in building end-to-end web applications using HTML, CSS, JavaScript, PHP, Laravel, React.js, and Vue.js for the frontend, as well as designing REST APIs, managing databases (PostgreSQL, MySQL, MongoDB, YugabyteDB), and implementing distributed systems using Python, Flask, and Docker for the backend. Focused on building functional, well-structured, and scalable digital solutions through various academic projects.

EDUCATION

UNIVERSITAS TEKNOLOGI DIGITAL INDONESIA

Bachelor of Informatics

Cumulative GPA: 3.67/4.00

Relevant Coursework: Algorithms & Data Structures, Data Visualization, Big Data, Operating Systems, Distributed Systems, UI/UX, Artificial Intelligence, Cybersecurity and Threat Intelligence.

Yogyakarta

Sep 2023 - Present

PROJECT EXPERIENCE

RAPID RESPONSE SYSTEM

Yogyakarta

Jun 2026

- Designed a mobile app prototype for UTDI's PPKS Task Force (Task Force for the Prevention and Handling of Sexual Violence) to speed up the case reporting and handling process.
- Created an interactive interface design in Figma with a navy and gold visual identity, complete with design tokens and component specifications for easy implementation by the development team.
- Structured the user navigation flow (user flow) to ensure the reporting process was fast, accessible, and maintained reporter confidentiality.

TRANSPORTATION GROWTH

Yogyakarta

Jun 2026

- Developed an interactive Streamlit-based dashboard to visualize motor vehicle growth data across 38 Indonesian provinces for 2023–2025 (BPS data source).
- Built various data visualizations (GeoJSON maps, bubble charts, and interactive filters) to facilitate cross-regional transportation trend analysis.
- Performed ongoing debugging and UI refinement (rendering fixes, sidebar, color contrast) and deployed the app to Streamlit Community Cloud.

DISTRIBUTED SYSTEM & REST API

Yogyakarta

May 2026

- Built a REST API using Python (Flask) connected to the distributed YugabyteDB database, run in a Docker container to support scalability and data replication.
- Implemented and explained the blockchain consensus mechanism (Ethereum Gasper) through technical diagrams as a case study in decentralized systems.
- Performed cross-platform debugging (Windows), including migrating the database driver (psycopg2 to pg8000) to resolve dependency compilation failures on Python 3.14.

ACADEMIC EXPERIENCE

TEACHING ASSISTANT – DATA STRUCTURES

Sep 2025

- Facilitated 37 students' understanding of complex algorithm concepts (Linked List, Stack, Queue, Tree) using Java.
- Translated abstract programming logic concepts into simple, visual explanations for easier student comprehension.
- Accurately managed student grading and attendance records using Spreadsheets and the E-learning platform.

TEACHING ASSISTANT – PROGRAMMING PARADIGMS

Sep 2025

- Assisted in teaching advanced Java programming topics including recursion, sorting, searching, data structures, and algorithmic problem-solving.
- Supervised lab practicum sessions and evaluated students' practicum reports.
- Guided students in applying algorithmic thinking to solve real-world programming problems.

ADDITIONAL EXPERIENCE

INFORMATICS & COMPUTER STUDENT ACTIVITY UNIT

Yogyakarta

Organizational Advisory Board

Aug 2024 - Present

- Provided strategic input to the organization's leadership on work program planning and decision-making.
- Evaluated the execution of organizational activities to ensure alignment with the organization's vision, mission, and rules.
- Provided input and recommendations to improve the effectiveness of work programs and organizational governance.
- Supported the organization's continuity through mentoring, coordination, and communication with leadership and members.

MASTER of CEREMONY

Yogyakarta

Universitas Teknologi Digital Indonesia

- Served as Master of Ceremony for various official campus events, including Graduation, Dies Natalis (Founding Anniversary), and PESONA (Study Introduction and Campus Orientation).
- Led event proceedings with effective communication, strong time management, and coordination with the committee and speakers.
- Demonstrated public speaking, event management, and adaptability in handling various situations during events.

EVENT DIVISION STAFF – UNIVERSITY GRADUATION

Yogyakarta

Universitas Teknologi Digital Indonesia

Oct 2025

- Participated in the planning and execution of the University Graduation ceremony series.
- Coordinated with cross-divisional committees to ensure the entire agenda ran on schedule.
- Managed technical readiness and event flow to support a smooth graduation ceremony.
- Helped resolve operational issues quickly and effectively during the event.

SKILLS

Programming Languages: Java, JavaScript, PHP, Python, CSS, HTML

Framework: Laravel, Vue.js, React.js, Streamlit

Database: PostgreSQL, MongoDB, MySQL

Data Analytic: Pandas, NumPy, Tableau, Power BI, Google Colab

Tools & Platforms: Git, Docker, Postman, Visual Studio Code, Figma, GitHub, WSL

Soft Skills: Problem Solving, Public Speaking, Teamwork, Time Management, Communication

Languages: Indonesian (Native), English (Intermediate)

Certifications:

- Web Programming Fundamentals – Dicoding | 2025
- Introduction to Information Security - Cyber Academy | 2026
- Panelist Youth Techbridge 2026: Developing Digital Leaders Across Borders – KKTU | 2026